

AI PRODUCT MANAGER / AI APPLICATIONS

ZHANG LEI | AUDIO GENERATION / MULTIMODAL AI / PRODUCT DELIVERY

EDUCATION

City University of Hong Kong (Dongguan)
MSc in Artificial Intelligence | 2026 - 2028
QS World University Ranking: 52

Wenzhou-Kean University
BSc in Mathematics and Applied Mathematics | 2022 - 2026
Minor in Computer Science

PROJECT

LLaVA-GI | Medical Visual Question Answering

- Trained a gastroenterology vision-language model on Kvasir-VQA using LLaVA, supporting image-and-text question answering; achieved 0.88 ROUGE-Precision on the test set.
- Independently completed Linux deployment, training, and preprocessing; resolved reproduction issues, submitted an upstream Issue, and tested robustness across models.

github.com/Lareina2441/LLaVA-GI

SKILLS

Programming
Python (proficient), Java, SQL

Frameworks and Tools
FastAPI, PyTorch, OpenCV, Transformers, MySQL, Linux, MATLAB, LaTeX, SPSS

AI and Data
TTS, LoRA / PEFT, multimodal models, LLM APIs, prompt engineering, SSE streaming, NLP, sentiment analysis

LANGUAGE TESTS

TOEFL 101 (R30 / L30 / S20 / W21)
GRE 325 (V155 / Q170 / AW3.5)

EXPERIENCE AND RESEARCH

Hangzhou Free-Form Technology | AI Product Manager Intern

2026.05 - 2026.09

- Data strategy:** researched Bilibili video characteristics and quality issues; designed collection and filtering SOPs with LLM-as-a-Judge, creating a 100,000-hour anime-audio corpus. Monitored and downloaded Douyin live audio through reverse-engineered interfaces, covering million-hour-scale data and expanding into TV drama and game dubbing.
- Data pipeline:** built volume normalization, VAD segmentation, alignment, ASR, speaker clustering, and annotation workflows to provide structured training data for audio-generation models.
- Model feature:** improved LoRA mixing through literature research; used independent autoregressive LoRAs to reduce cross-LoRA contamination and enable continuous male-female voice morphing.
- Evaluation:** summarized dataset characteristics through broad listening and manual records, then combined LLM synthesis with business goals to design model evaluation sets.

Shenzhen Shuyuxinyuan | AI Application Intern

2026.01 - 2026.03

- AI application delivery:** contributed to product design and front-end development for a therapeutic AI app, using Lovable, Figma, React, TypeScript, and Ant Design.
- Backend and data:** designed FastAPI diary-management services with multimodal parsing and persistent storage on Alibaba Cloud OSS / RDS and PostgreSQL.
- Dialogue and RAG:** designed selective history retrieval and context assembly, streamed LLM responses through SSE, and persisted complete conversations.
- Personalized voice:** generated tailored meditation scripts from diary and conversation context, then synthesized dedicated guidance audio with TTS.
- Delivery:** implemented sensitive-word filtering and other safety controls, produced API and deployment documentation, and completed Linux server deployment.

Tsinghua University Future Laboratory | Research Intern

2024.12 - 2025.08

Real-time Facial Expression and Negative Emotion Detection

- Extracted 468 3D facial landmarks with MediaPipe Face Mesh and trained an eight-class emotion-recognition model using a PyTorch Transformer.
- Built a multithreaded OpenCV pipeline for live camera streams and a threshold-based decision mechanism for low-latency negative-emotion detection.

Poetry Generation and Emotional Expression

- Built a voice input -> ASR -> ChatGPT poetry generation -> TTS reading loop for an older-adult AI interaction study, with complete setup and API documentation.